INCH-POUND

A-A-54016
19 March 1993

COMMERCIAL ITEM DESCRIPTION

REFRIGERATOR, SOLID STATE BIOLOGICALS

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a portable solid state biologicals refrigerator.

Salient characteristics:

The refrigerator shall be capable of maintaining operating temperatures between 33.8-42.8°, inclusive within all areas of the storage compartment. The refrigerator shall be capable of cooling the minimum payload of whole blood units described below from 50°F to 43°F, at an ambient temperature of 81°F, within seven hours. The refrigerator shall be capable of maintaining the payload temperature below 10°C following power loss at 77°F ambient temperature for a minimum of six hours providing the appropriate 34°F-43°F storage temperature existed prior to power failure. It shall be suited for use in all rugged field environments within temperatures ranging from -25°F through 120°F, inclusive.

The unit shall pull down to 6°C within 7 hours at an ambient temperature of 27.2°C (after being precooled to 6°C, with the full load of blood pre-cooled to 10°C).

The refrigerator with all accessories in place and containing a minimum payload of 30 whole blood units of 450 ml each or 50 packed red blood cell units of 270 ml each or the equivalent of either of the previous shall not exceed 125 pounds in weight and 8.5 cubic feet in external volume. Outside dimensions of unit shall not exceed 22.0 inches in depth or 32.0 inches in width (length). Volumes and dimensions do not include recording thermometer. Legs are not required on the refrigerator.

AMSC N/A FSC 4110

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The unit's internal storage capacity shall be capable of storing the minimum payload as described above. It shall have an easily removable basket(s) to facilitate loading and unloading.

The refrigerator shall be equipped with a temperature indicator, and an audible and visual alarm system which will alert the operator that an abnormal temperature condition exists or existed. The temperature indicator and alarm system shall be flush mounted as an integral part of the unit. The alarm shall be activated when the temperature falls outside the acceptable $34^{o}F$-$43^{o}F$ temperature range or in the event of electrical power loss. The alarm shall have a minimum dB rating of 60. The power source for the alarm system shall be separate from that of the refrigerator and shall incorporate a continually recharging battery power source or utilize a standard commercially available battery in the Federal Supply System. The capability to measure battery voltage or indicate low battery voltage shall be included in the alarm system.

The unit shall be certified by an independent testing laboratory to be in compliance with UL 471, Section 49, Rain Test; Section 37, Leakage Current Test; and Section 44, Dielectric Voltage Withstand Test.

The unit shall be supplied with a recording thermometer to provide a continuous hard copy recording of at least one week of storage compartment temperatures under non-transit conditions. The recording thermometer shall record all temperatures between minus $10^{o}C$ to plus $20^{o}C$, in increments of $2^{o}C$ as a minimum. The recording thermometer shall be supplied as an integral component of the thermometer alarm system or as a separate accessory item (packaged with the refrigerator). The maximum interval between points of record shall be 5 minutes. If the temperature recorder is supplied as an accessory, it shall have a carrying handle.

The unit shall be capable of operating from 110/220 VAC, 50/60 Hz and 12/24/28 VDC power supplies. AC and DC input power circuits shall be protected via circuit breakers. Control and power circuits shall be isolated from input line power circuit.

A minimum of 2 carrying handles shall be affixed to the refrigerator to facilitate loading and unloading the unit from vehicles and carrying the unit to the operational site.

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The refrigerator shall be durably constructed to withstand normal field handling and vibration peculiar to wheeled vehicle movement over roads, trails, and cross-country.

The unit shall be nonreflective "olive drab" in color. The units are not subject to biological and chemical warfare cleaning.

Workmanship. The solid state biologicals refrigerator shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One refrigerator, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be performed on a first article sample, as follows, to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

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1. Examine the unit for removable storage racks. Load the unit with thirty blood bags of 450 ml each. Examine the storage racks for bending or breaking.

2. Weigh the unit loaded with thirty blood bags of 450 ml each. The loaded unit shall weigh no more than 125 pounds.

3. Measure the external volume of the unit. The unit shall have an external volume of 8.5 cubic feet.

4. Examine the unit for power requirements. The unit shall operate on all of the following power sources:

120 VAC, 50 Hz	220 VAC, 60 Hz
120 VAC, 60 Hz	12 VDC
220 VAC, 50 Hz	24 VDC

5. Load the unit with 30 blood bags of 450 ml each filled with fluid. Precondition the unit to approximately 39°F. Vary the ambient temperature as follows and record the internal temperature via thermocouple each hour. The unit internal temperature shall stay within the range 34°F to 43°F. Power the unit from 120 VAC, 60 Hz.

Time	Ambient (°F)	Time	Ambient (°F)
0600	90	1900	114
0700	91	2000	108
0800	95	2100	105
0900	101	2200	102
1000	106	2300	100
1100	110	2400	98
1200	112	0100	95
1300	116	0200	94
1400	118	0300	93
1500	119	0400	92
1600	120	0500	91
1700	119	0600	90
1800	118		

6. Precondition the loaded unit to within the 34°F to 43°F range. Remove power from the unit. The internal temperature of the unit shall not exceed 50°F for a minimum of 6 hours with an ambient temperature of approximately 76°F.

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7. Load the refrigerator to specified capacity with blood bags filled with blood simulant. Operate the refrigerator until the blood simulant is within the 1°C to 6°C temperature range. Secure the refrigerator to a vibration platform. Excite the refrigerator at a 10 to 50 Hz frequency, with power spectral density of 0.15 g^2/Hz for 20 minutes. Upon completion of excitation, examine refrigerator for satisfactory operation.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. Failure of one unit to pass inspection shall result in failure of the lot. The unit of product for examination purposes shall be one solid state biologicals refrigerator as specified in this description.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and

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regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

4110-01-287-7111

MILITARY INTERESTS:

Custodians:

Air Force - 03
Army - MD
Navy - MS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA-MPQAS
USPHS
VA - OSS

Project Number: 4110-0432

Location: REFRIG17.DOC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-54016	2. DOCUMENT DATE (YYMMDD) 19 March 1993

3. DOCUMENT TITLE
REFRIGERATOR, SOLID STATE BIOLOGICALS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial: AC 215-737-2870 (2) AUTOVON: AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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